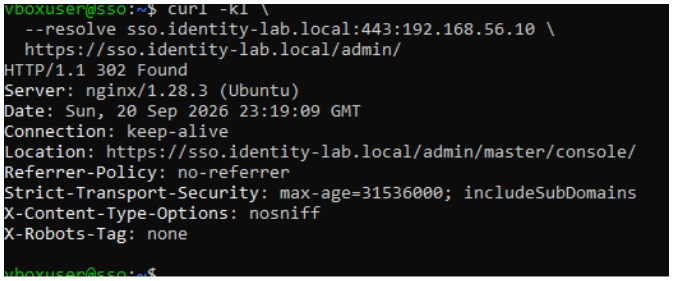
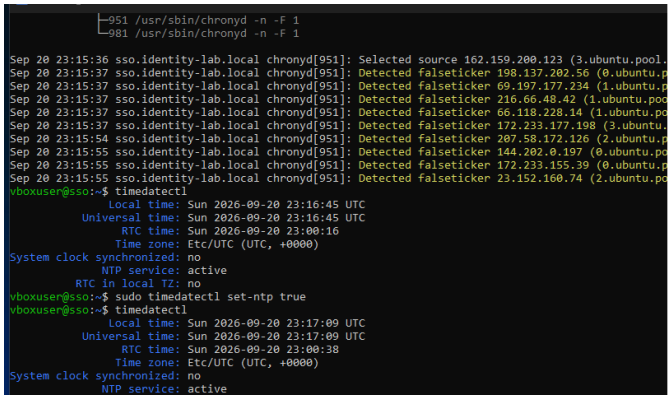
Screenshot 01 — Service Availability

The Docker containers and web endpoints were checked to verify the availability of the laboratory environment. The HTTP responses confirmed that Keycloak and Grafana were accessible.



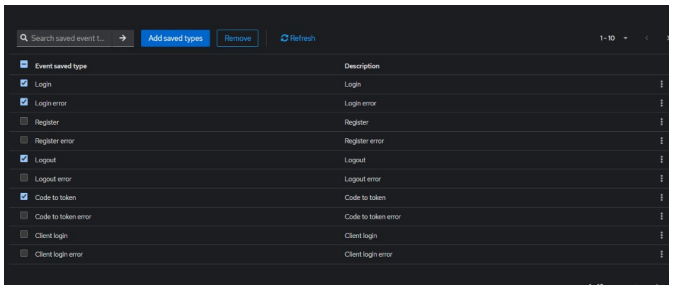
Screenshot 02 — Time Synchronization Check

The system time configuration and available time synchronization services were inspected. The `systemd-timesyncd` service was not found, so an alternative synchronization service required verification.



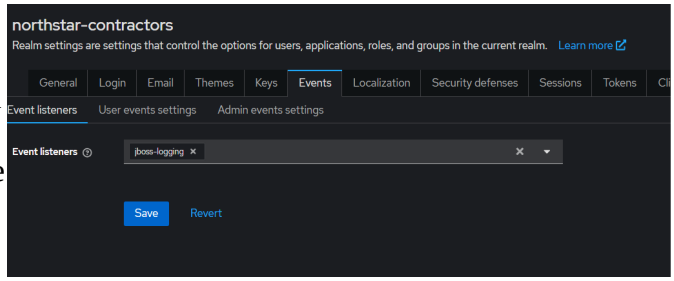
Screenshot 03 — User Event Settings

User event storage was enabled for the `northstar-contractors` realm with a retention period of one day. The selected event types included LOGIN, LOGIN_ERROR, LOGOUT, CODE_TO_TOKEN, UPDATE_PROFILE, and UPDATE_EMAIL.



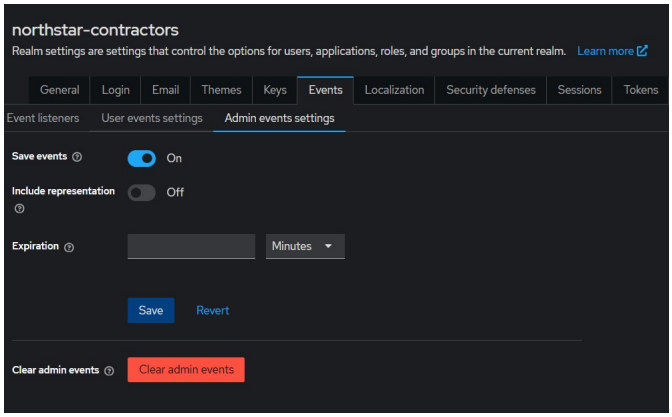
Screenshot 04 — Logging Event Listener

The `jboss-logging` listener was configured to send Keycloak events to the server log. This provides an additional source of audit information alongside the events stored in the database.

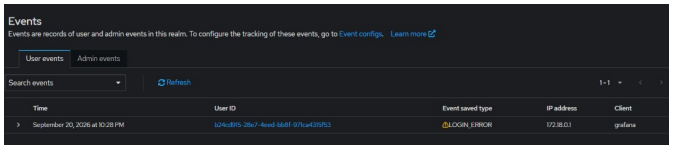


Screenshot 05 — Admin Event Settings

Administrative event storage was enabled to record changes made through the Admin Console and Admin REST API. The “Include representation” option was disabled to avoid storing complete request representations.

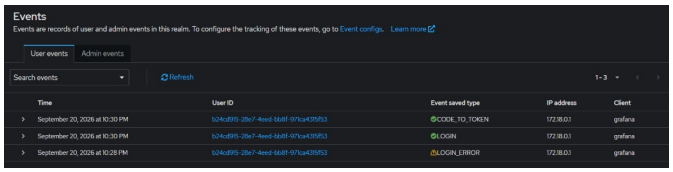


Screenshot 06 — Failed Login Event



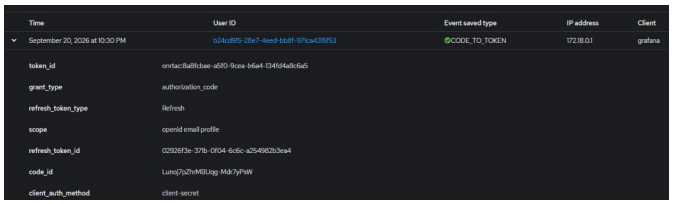
A single login attempt with an incorrect password generated a LOGIN_ERROR event. The event details were examined to identify the client, user information, source IP address, and recorded error.

Screenshot 07 — Successful Login Event



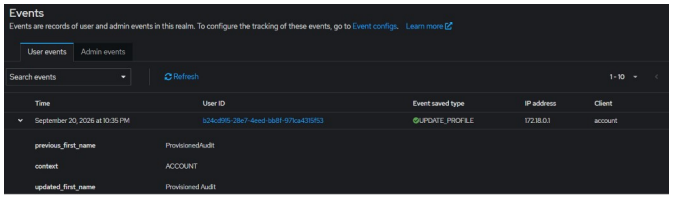
A successful sign-in to Grafana through Keycloak generated a LOGIN event. The timestamp, user identifier, client, and IP address were inspected to establish the context of the authentication.

Screenshot 08 — Authorization Code Exchange



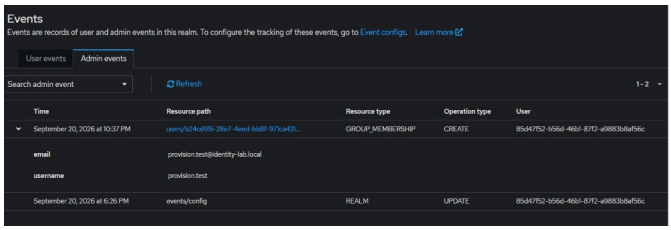
The CODE_TO_TOKEN event recorded Grafana's exchange of an authorization code for tokens. This event was correlated with the successful login to trace the OpenID Connect authentication flow.

Screenshot 10 — User Profile Update



The test user temporarily changed their first name through the Keycloak Account Console, generating an UPDATE_PROFILE event. The original value was restored after the audit record was inspected.

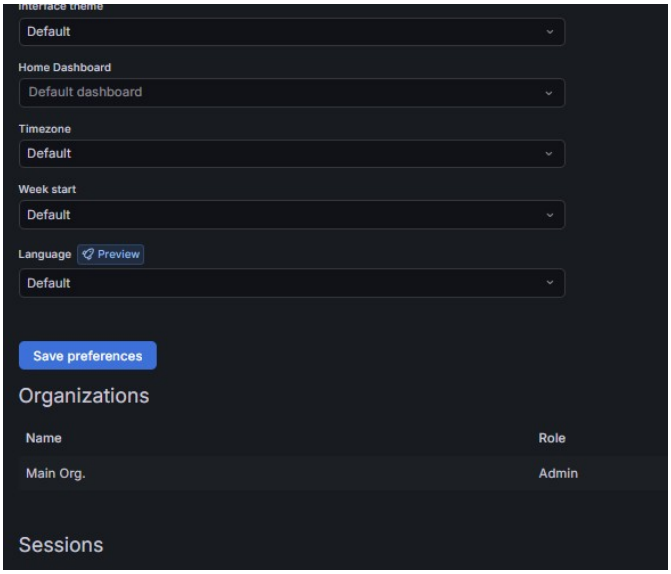
Screenshot 11 — JIT Group Membership Granted



The test user was added to the `grafana-jit-admins` group to grant temporary elevated access. The corresponding administrative event recorded the administrator's identity, timestamp, operation type, and resource path.

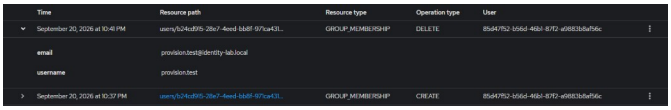
Screenshot 12 — Elevated Grafana Role

After a new sign-in, Grafana displayed the Admin role for the test user. This confirmed that the JIT group membership was reflected in the user's application permissions.



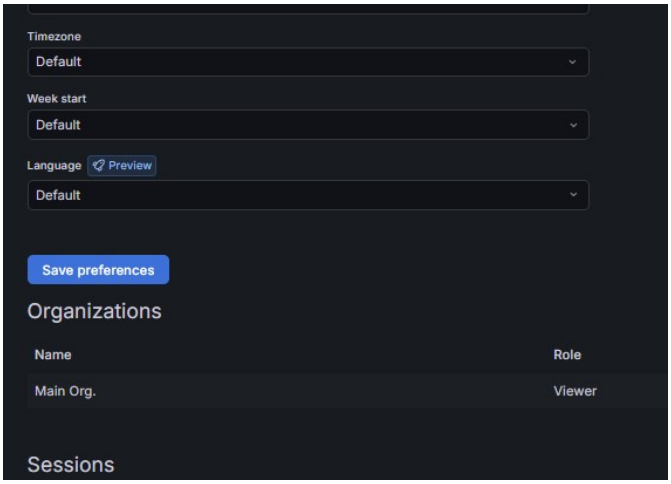
Screenshot 13 — JIT Group Membership Revoked

The test user's membership in `grafana-jit-admins` was removed. The administrative audit record was inspected to determine who revoked the membership and when the change occurred.



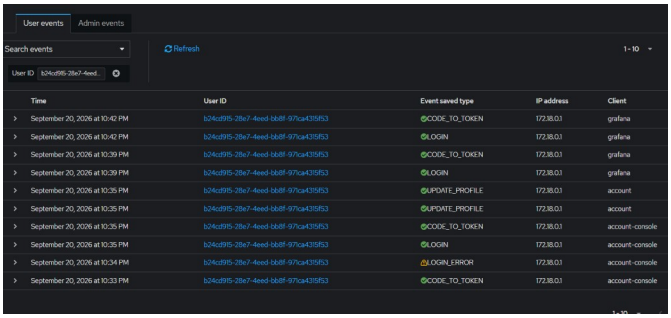
Screenshot 14 — Baseline Access Restored

The user signed in again after the JIT group membership was removed. Grafana displayed the Viewer role, confirming that the new session used the baseline permissions.



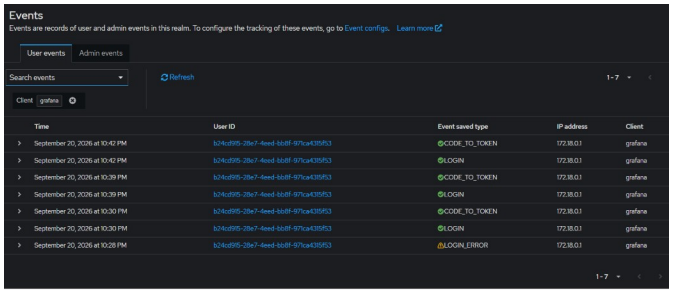
Screenshot 15 — Events Filtered by User

The user event log was filtered by the test user's identifier. This isolated the available records associated with that account for further analysis.



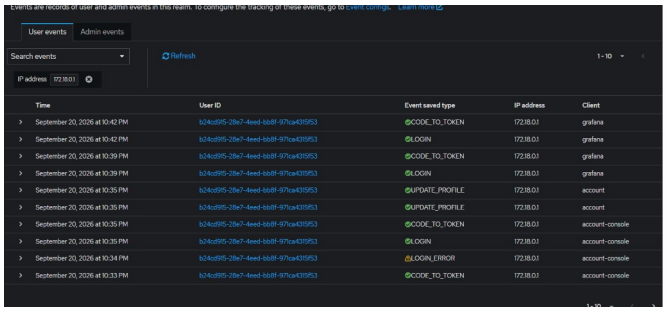
Screenshot 16 — Events Filtered by Client

The event log was filtered by the `grafana` client. The results isolated authentication-related events associated with Grafana.



Screenshot 17 — Events Filtered by IP Address and Date

Events were filtered by the recorded source IP address and the relevant date range. This narrowed the investigation to activity associated with that address during the laboratory exercise.



Screenshot 18 — Administrative Event Filtering

Administrative events were filtered to locate the JIT membership changes. The grant and removal records were compared using their timestamps, administrator information, and resource paths.

